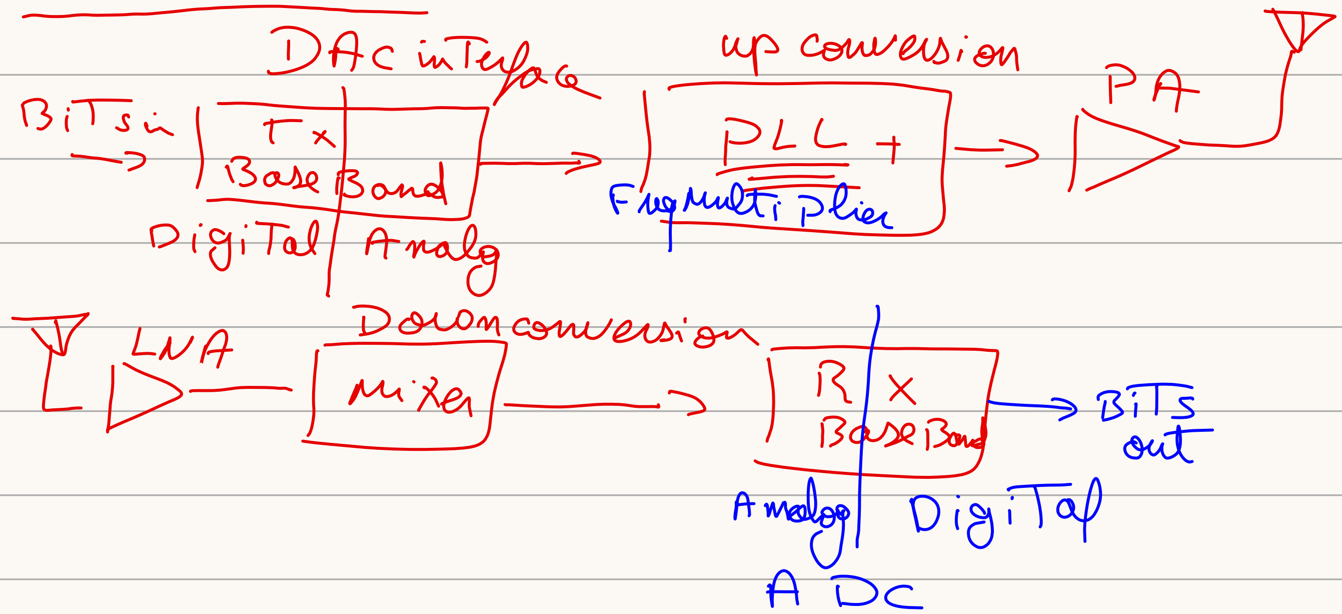
RF-lecture-1 => university of IDAHO

RF - Transceivers:



Antenna Basic's:

Antenna match energy on a Transmission line To a Free wave propagating in space

↳ has spec's like input impedance eff η, gain & Directivity

antenna with high gain has high Directivity

↳ omni directional antenna has low gain Ex: => Light Bulb

↳ Antenna with high gain => Ex: Flash light

Efficient antenna has Dimensions roughly equal To λ #

Friis propogation law:

$$P_r = G_t \cdot \frac{P_T}{4\pi r^2} \cdot A_{er}$$

$A_{er} \to \frac{\lambda^2}{4\pi} G_r$

$A_{er} \to$ Effective apeture area of The Rx

Radio Transceiver :- isolation :-

* Tx & Rx Don'T operate @ The same Time
  - → Frequency Division multiplixing
  - → Time " "
  - → off chip filters / Circulator / isolators

⟹ STill if Tx & Rx work simultisly ⟹ The isn'T enough ⟹ leackage To Rx ⟹ we could some circuit To cancel leackage.